Where Do Bacteria Live?

Aim: to take bacterial swabs and determine conditions for bacterial growth.

Equipment:

- Petri dish with nutrient agar
- Sterile cotton swabs
- Coloured permanent marker pens
- Ruler
- Clear sticky tape
- Incubator
- Location class list

Safety Information

Cultures – possible bacterial infection risk

Handle sealed Petri dishes only.
Dispose of in biohazard marked bags. Wash hands after handling.

Glassware – risk of cuts if broken

Apply pressure to wounds and inform teacher immediately.

Method:

Step 1: Your teacher will assign you a location from where to collect your sample.

Step 2: Without removing the lid, turn your Petri dish upside down and label the perimeter with your name and the date.

Step 3: Using a ruler, carefully draw a line across the middle of the Petri dish to divide it into two halves. On one half, write the name of the location from where you will collect your sample, and label the other half as 'control'.

Step 4: Remove the swab from the packaging, taking care not to touch the end. Gently rub the swab on the assigned location.

Step 5: Quickly but carefully open the lid of the Petri dish – just wide enough to fit the swab in – and rub the swab over the agar in a zig-zag pattern in the half labelled with your location. Take care not to mark the agar, or to rub onto the half marked as control.



Step 6: Replace the petri dish lid and secure on four sides with sticky tape. Leave some gaps to prevent anaerobic bacteria growing.

Step 7: Place your Petri dish, upside down, into the incubator. This helps to prevent condensation occurring, and allows easier identification of the plates when checking them.

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Step 8: Examine your Petri dish again after one day, two days and then five days. Record all observations on your result sheet.

Step 9: Your teacher may also conduct a control* investigation to measure bacteria from the air.

This could be done by simply leaving a Petri dish of agar open and exposed for a specified time length e.g. one hour, then sealing and incubating with the other Petri dishes.

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Location Class List

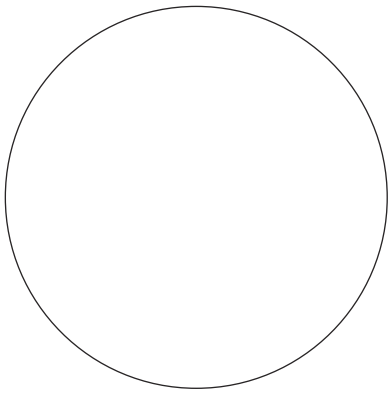
Teacher note: Suggested are several locations and space to add in your own. You may decide for students to investigate a location each or to work in pairs, depending on your class size. We recommend assigning the students to a location and them displaying the list to the class.

	Location description	Students investigation this area
1	Classroom desk	
2	Lab sink plug hole	
3	Lab sink tap handle	
4	Lab sink tap spout	
5	Classroom door handle	
6	Classroom light switch	
7	Toilet seat	
8	Toilet flush handle	
9	Toilet sink tap	
10	Toilet door handle	
11	Toilet hand dryer nozzle	
12	Mobile phone screen	
13	Projector remote	
14	Teacher's computer keyboard	
15	Teacher's computer mouse	
16	Inside of school bag	
17	Inside of shoe	
18	Your pencil	
19	Your exercise book	
20	Drinks bottle spout	
21	Classroom floor	
22		
23		
24		
25		

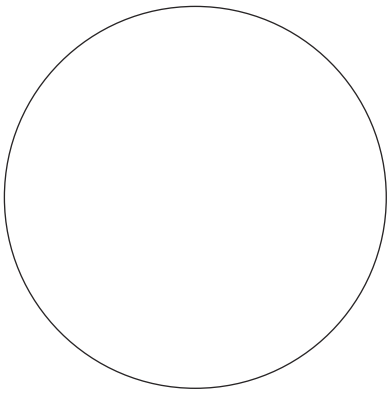
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Results

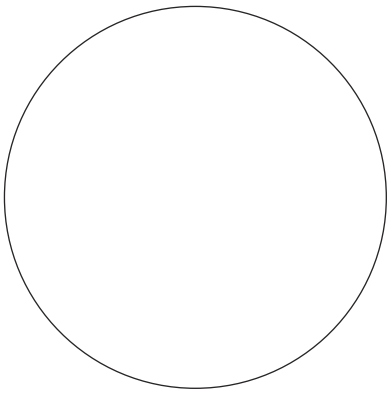
Draw the bacterial colonies (if any) which have grown on your agar plate.



After 1 day



After 2 days



After 5 days

Count the bacterial colonies that have grown on the agar and complete the table below.

Tips for counting the colonies:

- Dot each colony with a coloured marker as you count.
- Use a different colour each time you examine to count the additional colonies only. Add this amount to the previous count.
- Place the Petri dish on a piece of graph paper and count the colonies row by row.
- If there are many colonies to count, divide the area into 2 using a marker pen, count the colonies in $\frac{1}{2}$ then multiply by 2 to find the total.

Location:	
Day	Number of colonies

Compare your results with the other locations. Your teacher may ask you to add your results to a collective data table.

You may be given the opportunity to take photographs as a means of recording and comparing the bacterial growth from difference locations.

Questions

1. Which location has the most number of colonies on day 5? Is this the location you expected?
2. Which location has the least number of colonies on day 5? Is this the location you expected?
3. Why is it important to use sterile nutrient agar for the investigation?
4. Describe the common environmental conditions where the highest bacterial growth was found. (Think about temperature, light, moisture etc.)
5. What does the control Petri dish show? Explain why.
6. What measures can you take to reduce your exposure to bacteria?

Answers

1. Which location has the most number of colonies on day 5? Is this the location you expected?
   This will depend on the locations used and the results the students have obtained.

2. Which location has the least number of colonies on day 5? Is this the location you expected?
   This will depend on the locations used and the results the students have obtained.

3. Why is it important to use sterile nutrient agar for the investigation?
   It ensures that the culture(s) which grow on the agar after incubation, are only the ones from the swab, and therefore ensure that the investigation is valid.

4. Describe the common environmental conditions where the highest bacterial growth was found. (Think about temperature, light, moisture etc.)
   Bacteria are likely to be found in abundance in areas which are moist, and contain traces of food for nutrition. Also high traffic items or locations, e.g. mobile phones and door handles may also show high levels of bacteria.

5. What does the control Petri dish show? Explain why.
   The control dish will likely show some growth of bacterial cultures. These cultures will have come from the air around the Petri dish whilst it was left open.

6. What measures can you take to reduce your exposure to bacteria?

   - **Wash your handles before and after eating or preparing foods, suing hot water and soap.**
   - **Bleaching and disinfecting worktops and surfaces regularly.**
   - **Ensuring that mouths and noses are covered, and then hands are washed after coughing and sneezing.**
   - **Drying and removing excess moisture from areas such as the bath tub, shower head and kitchen sink.**
   - **Other sensible suggestions.**